

Presentation Transcript

Title Slide

I analyzed the ER visits dataset to identify discrepancies between the diagnosis rates for MBN disorders across the four US regions over the period 2016 to 2022. Before I explain why I chose this topic, it is important to define key terms.

MBN Disorders

MBN disorders— or mental, behavioral, and neurodevelopmental disorders—are grouped together in the current edition of the World Health Organization’s International Classification of Diseases. They are characterized by “clinically significant disturbance in an individual's cognition, emotional regulation, or behaviour” (WHO, 2019).

As for the distribution of these disorders across the human population, MBN disorders exist wherever humans exist (WHO, 2001). According to the Journal of World Psychiatry, researchers have begun to identify the risk factors for these disorders not as primarily biological but as social. For example, “the incidence and prevalence of many psychiatric disorders differ by biological sex,” but these differences are not purely a result of biology but rather of socialization (Kirkbride et al., 2024). Personal experiences are better predictors of MBN diagnosis than biological identity. Therefore, MBN disorders may disproportionately affect populations which experience more hardships, such as minorities.

US Regions

The regions of the United States are defined by the United States Census Bureau as Northeast, Midwest, South, and West (United States Census Bureau, 2021). Although states are not grouped into these regions based on cultural similarity, these patterns do tend to appear as a result of proximity and the regions gain distinct identities.

South Statistics

Perhaps the most distinct of the regions is the South, which has a notoriously bad reputation for anything you could think of. Wellbeing Rankings from Social Indicators Research rank US states from 1 to 50 for a number of factors including health, social and economic, and physical environment, and the average state ranking for Southern states was 29.42 (Blanchflower & Bryson, 2024). This ranking was the closest to 50 of the four

regions, indicating the lower quality of life in the Southern United States compared to other regions. If more people are living a worse quality life in the Southern United States, it stands to reason that more people in that region may be suffering from MBN disorders, which brings me to my research question and hypothesis.

Hypothesis / Research Question / Goal

I wanted to answer the question: How do emergency room mental, behavioral, and neurodevelopment diagnosis rates vary across regions (Northeast, Midwest, South, and West) of the United States? But instead of predicting that the Southern United States would have more diagnoses of MBN disorders, I predicted that it would actually have less. I do that for this reason. A separate analysis of the state of mental health in America by Mental Health America ranks states from 1 to 50 by 17 measures, including mental health workforce availability (Reinert et al., 2025). The average ranking for states in the South was 35.38, which was the closest to 50 of the four regions, indicating that the Southern United States has the worst conditions for mental healthcare of the four regions. Why does this matter? My grandfather was a pathologist and he used to say “doctors diagnose what they can treat.” I think that’s partially true, but there is more truth in the statement that “doctors diagnose what they can recognize.” Healthcare professionals lacking mental health training will fail to diagnose mental health issues compared to those with mental health training, and research indicates that proper training is least common in the South (Korankye et al., 2025).

Additionally, mental health stigma is more concentrated in rural areas, and the Southern United States is composed of large rural areas. Mental health stigma, too, acts to prevent diagnosis (Ben-Zeev et al., 2010).

So, I wanted to focus on the South because I believed the results would be somewhat paradoxical to established research. Negative social determinants are known to increase development of MBN disorders, while stigma and lack of mental healthcare infrastructure are known to decrease diagnosis rates. These are the factors that led me to believe diagnosis rates might be lower in the Southern United States despite the potential higher concentration of individuals actually affected by MBN disorders. There are many factors that could explain unusually low diagnosis rates, but the social nature of MBN disorders makes this more likely.

My goal was to show that the average ER diagnosis rate of MBN disorders would be significantly lower in the South compared to other US Regions with the hope that these results may serve as a call-to-action for the improvement of mental healthcare in the Southern United States.

Methods / Procedures

To generate an answer to my research question, I first cleaned the data in R. The ER visits dataset had 6,776 observations of 12 variables over the period 2016 to 2022 in its original form. The complete dataset provides both emergency room visit rates and visit counts by primary diagnosis for a number of diagnoses and by reason for visit for a number of symptoms. The rates and visit counts for each symptom or diagnosis are also divided into multiple social categories such as race, age, and geographic location. For my purposes, I narrowed the set down to 28 observations of four variables, all visit rates for MBN diagnoses by category of US region.

After cleaning the data, I graphed the average diagnosis rate for MBN disorders per 1,000 civilians in each region with a bar graph. I used the standard error of the mean times two for error bars to show statistical significance. I also created a line graph to show how the diagnosis rates changed over the years by region.

Finally, I ran an Analysis of Variance (ANOVA) to determine the significance of my data.

Results

This is the bar graph I created in R. As you can see, the South has the lowest average diagnosis rate for MBN disorders, which is what I expected, but ANOVA is required to determine the significance of these results.

Additional Graph

Additionally, the line graph that I created in R shows that the yearly ER diagnosis rates for MBN disorders were consistently—6 of 7 years—lower than the other regions. You can also see how the other regions compare to each other.

Statistical Analysis

The p-value from my ANOVA was 0.000154, which is much less than 0.05, therefore signifying that there is significant evidence that the differences in diagnosis rates by region are statistically significant rather than a result of random chance.

As part of my ANOVA, I also conducted a Tukey Honestly Significant Difference (HSD) test to determine the relationship between each of the regions. As you can see, differences in the average ER diagnosis rates between the West vs. Northeast and West vs. Midwest were not significant, as expected. Conversely, the differences in the average diagnosis rates between the South and West and South and Northeast were significant, as I predicted.

Unexpectedly, the differences in the average diagnosis rates between the South and Midwest were also insignificant. Additionally, the differences in the average diagnosis rates between the Northeast and Midwest were significant, which I also did not predict.

Discussion / Conclusions / Future Directions

According to the low p-value from ANOVA, US regions do vary significantly in ER diagnosis rates for mental behavioral, and neurodevelopmental disorders, however not all comparisons of diagnosis rates for MBN disorders across US Regions are significant. A drawback of this analysis is the small sample size. Each region only had seven observations—one per year from 2016 to 2022. If the surveillance period had been longer, there would be more testable data, which may produce different results.

The Southern United States showed significant differences from two of the other three US regions rather than all three. Although the southern United States did have the lowest average emergency room diagnosis rate for mental, behavioral, and neurodevelopmental disorders, these results were only partially significant. The comparison of the South and Midwest was not significant, and this may be because the South and Midwest have one large thing in common: rurality. Both the Midwest and South have large portions of their regional populations concentrated in rural areas. As established, mental health infrastructure and stigma are both worse in rural areas, decreasing the ability of inhabitants to receive diagnoses. In this way, the insignificance of the comparison is, itself, significant.

Ultimately, I did succeed in my goal of highlighting the effects of decreased quality of mental healthcare in the Southern United States, but I was also able to identify a larger pattern. Rural areas suffer from unique ailments, and the South is not the exclusive hub for rurality. It is my hope that these results not only draw attention to the decreased quality of mental healthcare in the Southern United States but also of rural areas in general. Moving forward, a comparison of rural and urban areas may be better than focusing solely on region. Rather than subscribing to the narrative that the South is automatically inferior because of its reputation, we should identify and study the larger structural forces that produce the infamous disparities associated with the region.

Thank you.

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